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Assignment-7

[Java/StudentRecordManager.java at main · diyaa-13/Java]

```java
import java.io.*;
import java.util.*;

public class StudentRecordManager {

    static final String FILE_NAME = "Students.csv";
    static Scanner sc = new Scanner(System.in);

    // ── Create File ──────────────────────────────────────
    static void createFile() {
        try (BufferedWriter bw = new BufferedWriter(new FileWriter(FILE_NAME))) {

            bw.write(str: "studentId,name,branch,marks1,marks2,marks3,marks4,marks5,percentage");
            bw.newLine();

            bw.write(str: "1,Aditya,CSE,80,85,90,88,96,0");
            bw.newLine();

            bw.write(str: "2,Riya,IT,75,70,80,76,87,0");
            bw.newLine();

            System.out.println(x: "File created successfully.");

        } catch (IOException e) {
            System.out.println("[ERROR] " + e.getMessage());
        }
    }

    // ── Add Records ──────────────────────────────────────
    static void addRecords() {
        try (BufferedWriter bw = new BufferedWriter(new FileWriter(FILE_NAME, append: true))) {

            for (int i = 0; i < 3; i++) {
                System.out.print(s: "ID: ");
                int id = Integer.parseInt(sc.nextLine());

                System.out.print(s: "Name: ");
                String name = sc.nextLine();
```

```java
            System.out.print(s: "Branch: ");
            String branch = sc.nextLine();

            System.out.print(s: "Marks1: ");
            int m1 = Integer.parseInt(sc.nextLine());

            System.out.print(s: "Marks2: ");
            int m2 = Integer.parseInt(sc.nextLine());

            System.out.print(s: "Marks3: ");
            int m3 = Integer.parseInt(sc.nextLine());

            bw.write(id + "," + name + "," + branch + "," + m1 + "," + m2 + "," + m3 + ",0,0,0");
            bw.newLine();
        }

        System.out.println(x: "3 records added.");

    } catch (IOException | NumberFormatException e) {
```

```java
            System.out.print(s: "Branch: ");
            String branch = sc.nextLine();

            System.out.print(s: "Marks1: ");
            int m1 = Integer.parseInt(sc.nextLine());

            System.out.print(s: "Marks2: ");
            int m2 = Integer.parseInt(sc.nextLine());

            System.out.print(s: "Marks3: ");
            int m3 = Integer.parseInt(sc.nextLine());

            bw.write(id + "," + name + "," + branch + "," + m1 + "," + m2 + "," + m3 + ",0,0,0");
            bw.newLine();
        }

        System.out.println(x: "3 records added.");

    } catch (IOException | NumberFormatException e) {
        System.out.println("[ERROR] " + e.getMessage());
    }
}

// ── Display ──────────────────────────────────────────────
static void display() {
    try (BufferedReader br = new BufferedReader(new FileReader(FILE_NAME))) {
        String line;
        System.out.println(x: "\n--- Records ---");
        while ((line = br.readLine()) != null) {
            System.out.println(line);
        }
    } catch (IOException e) {
        System.out.println("[ERROR] " + e.getMessage());
    }
}
```

```java
// ── Update Marks4 & Marks5 ──────────────────────────────
static void updateMarks() {
    try {
        File file = new File(FILE_NAME);
        List<String> lines = new ArrayList<>();

        BufferedReader br = new BufferedReader(new FileReader(file));
        String line;
        lines.add(br.readLine()); // header

        while ((line = br.readLine()) != null) {
            String[] p = line.split(regex: ",");

            System.out.println("Updating for ID: " + p[0]);

            System.out.print(s: "Marks4: ");
            p[6] = sc.nextLine();

            System.out.print(s: "Marks5: ");
            p[7] = sc.nextLine();

            lines.add(String.join(delimiter: ",", p));
        }
        br.close();

        BufferedWriter bw = new BufferedWriter(new FileWriter(file));
        for (String l : lines) {
            bw.write(l);
            bw.newLine();
        }
        bw.close();

        System.out.println(x: "Marks updated.");

    } catch (IOException e) {
        System.out.println("[ERROR] " + e.getMessage());
```

```java
    }
}

// ── Calculate Percentage ──────────────────────────────────
static void calculatePercentage() {
    try {
        File file = new File(FILE_NAME);
        List<String> lines = new ArrayList<>();

        BufferedReader br = new BufferedReader(new FileReader(file));
        String line;
        lines.add(br.readLine());

        while ((line = br.readLine()) != null) {
            String[] p = line.split(regex: ",");

            double sum = 0;
            for (int i = 3; i <= 7; i++) {
                sum += Double.parseDouble(p[i]);
            }

            double percent = sum / 5;
            p[8] = String.format(format: "%.2f", percent);

            lines.add(String.join(delimiter: ",", p));
        }
        br.close();

        BufferedWriter bw = new BufferedWriter(new FileWriter(file));
        for (String l : lines) {
            bw.write(l);
            bw.newLine();
        }
        bw.close();

        System.out.println(x: "Percentage updated.");
```

```java
        } catch (IOException | NumberFormatException e) {
            System.out.println("[ERROR] " + e.getMessage());
        }
    }

    // ── Delete Record ──────────────────────────────────────────
    static void deleteRecord() {
        try {
            System.out.print(s: "Enter ID to delete: ");
            int id = Integer.parseInt(sc.nextLine());

            File file = new File(FILE_NAME);
            List<String> lines = new ArrayList<>();

            BufferedReader br = new BufferedReader(new FileReader(file));
            String line;
            lines.add(br.readLine());

            boolean found = false;

            while ((line = br.readLine()) != null) {
                String[] p = line.split(regex: ",");

                if (Integer.parseInt(p[0]) == id) {
                    found = true;
                } else {
                    lines.add(line);
                }
            }
            br.close();

            BufferedWriter bw = new BufferedWriter(new FileWriter(file));
            for (String l : lines) {
                bw.write(l);
                bw.newLine();
            }
```

```java
            bw.close();

            System.out.println(found ? "Deleted successfully." : "ID not found.");

        } catch (IOException | NumberFormatException e) {
            System.out.println("[ERROR] " + e.getMessage());
        }
    }

    // ── Exception Demo ──────────────────────────────────────
    static void exceptionDemo() {
        try {
            BufferedReader br = new BufferedReader(new FileReader(fileName: "wrongfile.csv"));
        } catch (IOException e) {
            System.out.println("[EXCEPTION DEMO] " + e.getMessage());
        }
    }

    // ── MAIN MENU ───────────────────────────────────────────
    Run | Debug
    public static void main(String[] args) {

        while (true) {
            System.out.println(x: "\n==== STUDENT CSV MENU ====");
            System.out.println(x: "1. Create File");
            System.out.println(x: "2. Add 3 Records");
            System.out.println(x: "3. Display Records");
            System.out.println(x: "4. Update Marks4 & Marks5");
            System.out.println(x: "5. Calculate Percentage");
            System.out.println(x: "6. Delete Record");
            System.out.println(x: "7. Exception Demo");
            System.out.println(x: "0. Exit");

            System.out.print(s: "Enter choice: ");
```

```java
            int ch;
            try {
                ch = Integer.parseInt(sc.nextLine());
            } catch (NumberFormatException e) {
                System.out.println(x: "Invalid input!");
                continue;
            }

            switch (ch) {
                case 1: createFile(); break;
                case 2: addRecords(); break;
                case 3: display(); break;
                case 4: updateMarks(); break;
                case 5: calculatePercentage(); break;
                case 6: deleteRecord(); break;
                case 7: exceptionDemo(); break;
                case 0:
                    System.out.println(x: "Exiting...");
                    sc.close();
                    return;
                default:
                    System.out.println(x: "Invalid choice!");
            }
        }
    }
}
```

<u>Output:</u>

```
PS C:\Users\vasum\Desktop\Diya\Java\Java>  & 'C:\D
' '-cp' 'C:\Users\vasum\AppData\Roaming\Code\User\

==== STUDENT CSV MENU ====
1. Create File
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 1
File created successfully.

==== STUDENT CSV MENU ====
1. Create File
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 2
ID: 101
Name: Diya
Branch: AIML
Marks1: 20
Marks2: 15
Marks3: 17
ID: 102
Name: Wani
```

```
3 records added.

==== STUDENT CSV MENU ====
1. Create File
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 3

--- Records ---
studentId,name,branch,marks1,marks2,marks3,marks4,marks5,percentage
1,Aditya,CSE,80,85,90,88,96,0
2,Riya,IT,75,70,80,76,87,0
101,Diya,AIML,20,15,17,0,0,0
102,Wani,AIML,20,20,19,0,0,0
103,XYZ,AIML,10,15,9,0,0,0

==== STUDENT CSV MENU ====
1. Create File
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 4
Updating for ID: 1
Marks4: 30
Marks5: 20
Updating for ID: 2
Marks4: 25
Marks5: 45
Updating for ID: 101
Marks4: 23
Marks5: 14
Updating for ID: 102
```

```
Marks4: 54
Marks5: 21
Updating for ID: 103
Marks4: 34
Marks5: 51
Marks updated.

==== STUDENT CSV MENU ====
1. Create File
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 5
Percentage updated.

==== STUDENT CSV MENU ====
1. Create File
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 5
Percentage updated.
```

```
Enter ID to delete: 103
Deleted successfully.

==== STUDENT CSV MENU ====
1. Create File
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 7
[EXCEPTION DEMO] wrongfile.csv (The system cannot find the file specified)

==== STUDENT CSV MENU ====
1. Create File
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 0
==== STUDENT CSV MENU ====
1. Create File
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 0
2. Add 3 Records
3. Display Records
4. Update Marks4 & Marks5
5. Calculate Percentage
6. Delete Record
7. Exception Demo
```

```
0. Exit
Enter choice: 0
6. Delete Record
7. Exception Demo
0. Exit
Enter choice: 0
Exiting...
```